

Tahreem Karim

Applied Data Scientist | Computational Biologist

Summary:

Former wet lab scientist turned data scientist with computational biology expertise and extensive experience in experimental research. Focused on bridging wet lab and computational workflows by transforming underutilized/archived simulators into intuitive tools that scientists actually use. Currently building custom data visualizations, molecular simulations, & stakeholder facing tools.

Work experience:

Consulting Computational Biologist

Contract

(Nonprofit research initiative)
Jan 2026 – Present

- Consulting practice, delivering image analysis, statistical methodology, and data visualization systems for research data to academic, private sector, and non-profit clients.
- Current engagement: Brown University's Warren Alpert Medical School – visualizing aging biomarkers and statistics in cancer pathology specimens.
- Delivered prior statistical methodology consulting (NDA-governed) for stroke and speech-language research with NYU's Department of Neuropsychology, & for neurodevelopmental disorder research with Harvard/VA South Shore.

Intern:

Computational Biology

Seanome

(Nonprofit research initiative)
Dec 2024 – Nov 2025

- Contributed to exploratory computational biology research leveraging ocean biodiversity datasets to inform protein annotation & polymer modeling.
- Designed resource database pipeline & documentation templates.
- Evaluated protein annotation literature for sensitivity, interpretability, & applicability through journal clubs.

Research Associate / Lab Manager

Chan Zuckerberg Biohub
Apr 2021 – Aug 2022

- Built & maintained **3D image – machine learning algorithm distinguishing 2,200+ GFP tagged cell lines**. Designed drug perturbation assay to output microscopy readouts of cellular responses/protein localization.
- Designed, cloned, & presented a **fluorescent cellular barcoding system** enabling multiplexed confocal imaging, reducing workload by 75%.
- Coordinated collaboration across microscopy, molecular biology, and bioengineering teams to implement scalable wet lab automation.
- Maintained lab instruments and robots. Managed compliance for 3+ teams.

Education:

University of Michigan – MS, Data Science

Data Visualization, Machine Learning,
Big Data Systems (in progress)

University of California, Berkeley – BS, Molecular & Cell Biology

Immunology & Pathogenesis
Minor: Philosophy

Selected Certifications

Neural Networks and Deep Learning – DeepLearning.AI
Algorithms – Stanford
Python for Data Science – University of Michigan

Core Tools

Python · R · HTML · Unix/Linux command line · SQL · Pandas ·
Notion · NumPy · SciPy · Streamlit · Altair · Matplotlib · Seaborn ·
Git · AWS · Notion · FoldSeek · BLAST · UniProt · GenBank · Benchling

Statistical and Algorithmic Methods

OLS & logistic regression · Pearson coefficients · Bootstrapping ·
Retrieval-Augmented Generation (RAG) · Streaming Algos ·
Dynamic programming · Hash tables · Heaps · Qualitative inquiry

Molecular Biology Techniques

Cloning · Plasmid/primer design · PCR · qPCR · RT-PCR ·
Lentiviral delivery · Tissue/cell culture · Hybridoma generation ·
FACS sorting · Multicolor flow cytometry · Western blot · ELISA ·
Confocal microscopy · Robot development · Project ownership